Beyond Reduced-Order Control: Learning the Coupled Nine-DOF Nonlinear Dynamics of a Wave Energy Converter

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The University of Western Australia
Honours research with WaveX

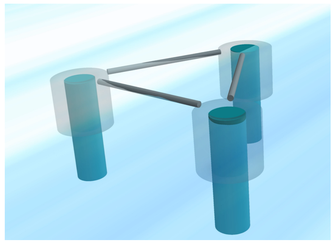
CSSE 50th anniversary

RL4XS 2026 · Oral Spotlight

THE CONTROL CHALLENGE

How hard do we resist the next wave to maximise power capture over time?

Ocean waves drive each D-Spar's relative motion. PTO damping sets the resistance and changes the resulting motion.



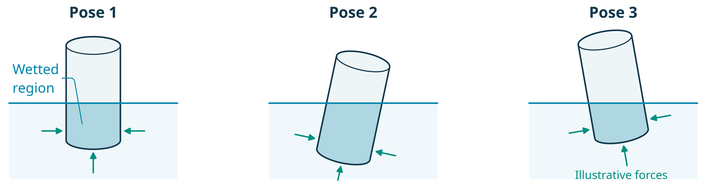
Each inner spar heaves independently relative to its outer spar. The rigid frame couples the platform: 6 platform DOFs + 3 relative-heave DOFs. Water-surface colour shows ocean-wave elevation.

CALCULATE THE KNOWN PHYSICS; LEARN THE NONLINEAR CORRECTION

Changing wetted geometry

$$(\mathbf{M} + \mathbf{A}_\infty)\dot{\mathbf{v}} = \mathbf{F}_{\text{known}} + \widehat{\boldsymbol{\phi}}_\theta(\mathbf{q}, \mathbf{v})$$

Known mechanics + learned nonlinear force correction

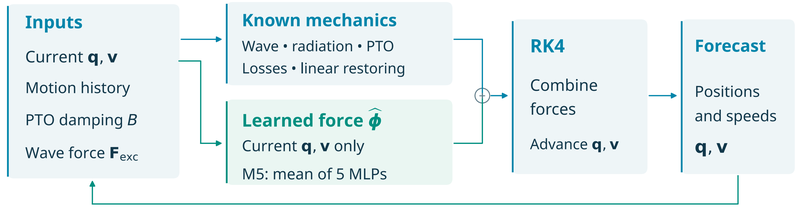


Pose changes submerged volume and centre of buoyancy.

Schematic geometry; reference waterline $z = 0$.

Grey-box recursive MLP rollout

Ocean-wave elevation → physical excitation + ridge correction
$\mathbf{F}_{\text{exc}}$ prepared once per forecast request

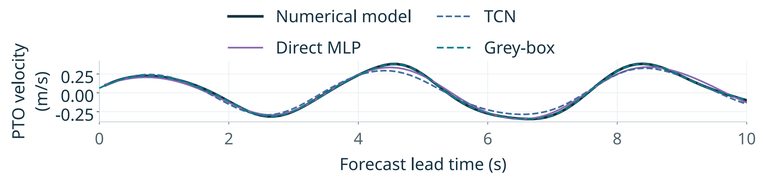


Feed back motion + radiation history

Repeat the force-and-integration step through the horizon.

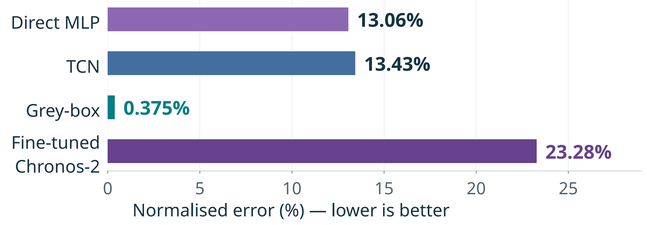
PREDICTING MOTION

10 s forecast example · B = 6,467 N·s/m



Motion prediction across the damping range

MLP / TCN / grey-box: 21 settings · 2k–100k N s/m

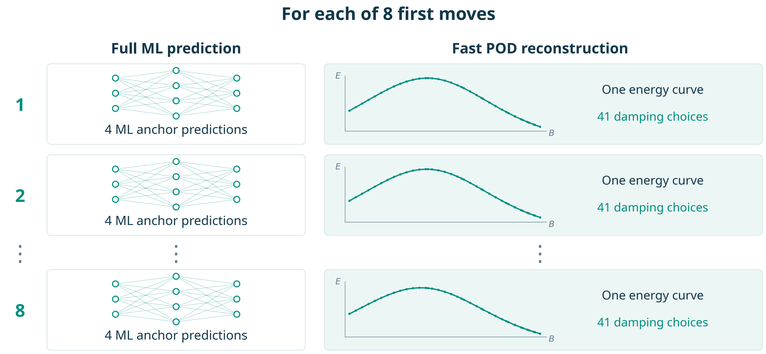


Provisional comparison · evaluation cohorts differ
Chronos: separate changing-damping validation

MLP/TCN: development waves 19/20; grey-box: model-held 21/22.
Chronos: separate validation wave 3. Matched data pending.

NONLINEAR PREDICTIVE CONTROL

Explore 8 × 41 candidate trajectories



8 × 41 = 328 candidate control sequences

Archived fitting example · predicted energy · shared scales, log B

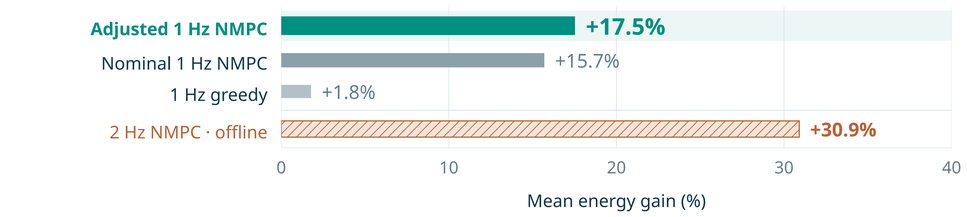
$$\max_{i,j} E_{ij} = \int_0^{10\,\text{s}} B_{ij}(t) \sum_{k=7}^{9} v_k^2(t)\,dt$$

Adjusted 1 Hz: execute A then 0.88B, 0.5 s each (bounded).
2 Hz: execute A for 0.5 s, then replan. See QR for full settings.

RESULTS IN NUMERICAL SIMULATION

+17.5%

mean mechanical energy adjusted 1 Hz NMPC



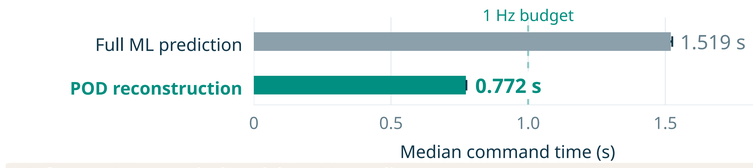
2 Hz is offline: 0.77 s median computation exceeds its 0.5 s budget.

Mean paired gain over fixed B = 5,000 N·s/m; 9 states × 4 waves.
300 s per wave; known future waves; combined descriptive cohort
(original two + fresh two waves per state).

Full ML prediction and POD

Reference nonlinear model shown separately

8 × 41 = 328 candidate trajectories



Reference NL numerical model: 40.84 s median

Separate earlier search: 41 screen actions + 42–49 plans · CPU / MATLAB

Current: 8 requests × 10; same initial POD screen; RTX 5000, 4 lanes.
Native: 8 requests × 3, CPU/MATLAB. Both: Hs = 0.75 m, Tp = 3 s.

Explore the project

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Paper, results & contact

reuben-trade.github.io/wavex/

Earlier CEED paper and updated results are clearly identified on the project page.

Scope: known ocean-wave preview; uniform PTO damping; 1 s application delay for both controllers; numerical validation. Next: causal forecasts and device trials. Mentors: Tim Green and Amal Jose Arackal.